

Natural Language Processing

Sequence-to-sequence Models and Attention Mechanisms

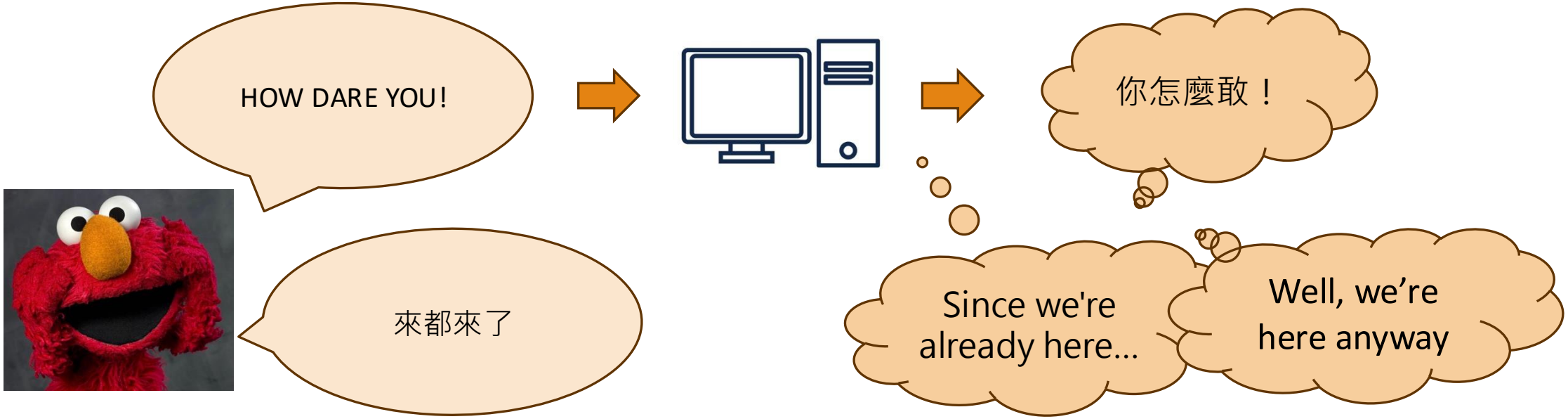
Syllabus of NLP course in University of Oslo, Norway

Aug 18	Introduction [YS+PL]	
Aug 25	Language models, tokenization [YS]	**Oblig 1**: LMs and text classification *Deadline*: Sep 11
Sep 1	Text classification with log. regression [YS]	
Sep 8	Sequence labelling [YS]	
Sep 15	Neural networks [PL]	**Oblig 2**: Sequence labelling and NNs *Deadline*: Oct 2
Sep 22	Vector semantics, word embeddings [YS]	
Sep 29	Transformers [YS]	
Oct 6	Large language models [PL]	**Oblig 3:** LLMs and RAG *Deadline:* Oct 23
Oct 13	Dialog systems I [PL]	
Oct 20	Dialog systems II, question answering [PL]	
Oct 28	Machine translation [YS]	**Oblig 4:** dialogue systems and MT *Deadline:* Nov 13
Nov 3	Speech processing [PL]	
Nov 10	Ethics I [PL]	
Nov 17	Ethics II, exam preparation [YS+PL]	**Exam:** Dec 18

Machine Translation

Machine translation plays a pivotal role in the development of NLP.

- Many advancements in NLP language models were initially motivated by the need to address translation challenges.



Challenges of Machine Translation

Unlike the other tasks like classification, the input and output of machine translation are **in different lengths**.



English
Short (3 tokens)

Chinese
(4 tokens)

Japanese
Long (12 tokens)

× We cannot just add an FFN at the end.

√ The hidden state should be utilized to encode the original sequence and it should be passed to the generation process.

Sequence to sequence Model

Sequence to Sequence (Seq2Seq) model

The core idea of the Seq2Seq model is to map **variable-length input sequences to variable-length output sequences**.

This flexibility enables its wide application across various tasks, such as:

- Machine Translation: Translating a sentence from one language to another.
- Text Summarization: Condensing long articles into shorter summaries.
- Dialogue Generation: Generating appropriate responses based on input contexts.

Sequential Models

Sequence models, also known as time series models, are a class of neural network architectures designed to model sequences of data.

- **RNN**
- **LSTM**

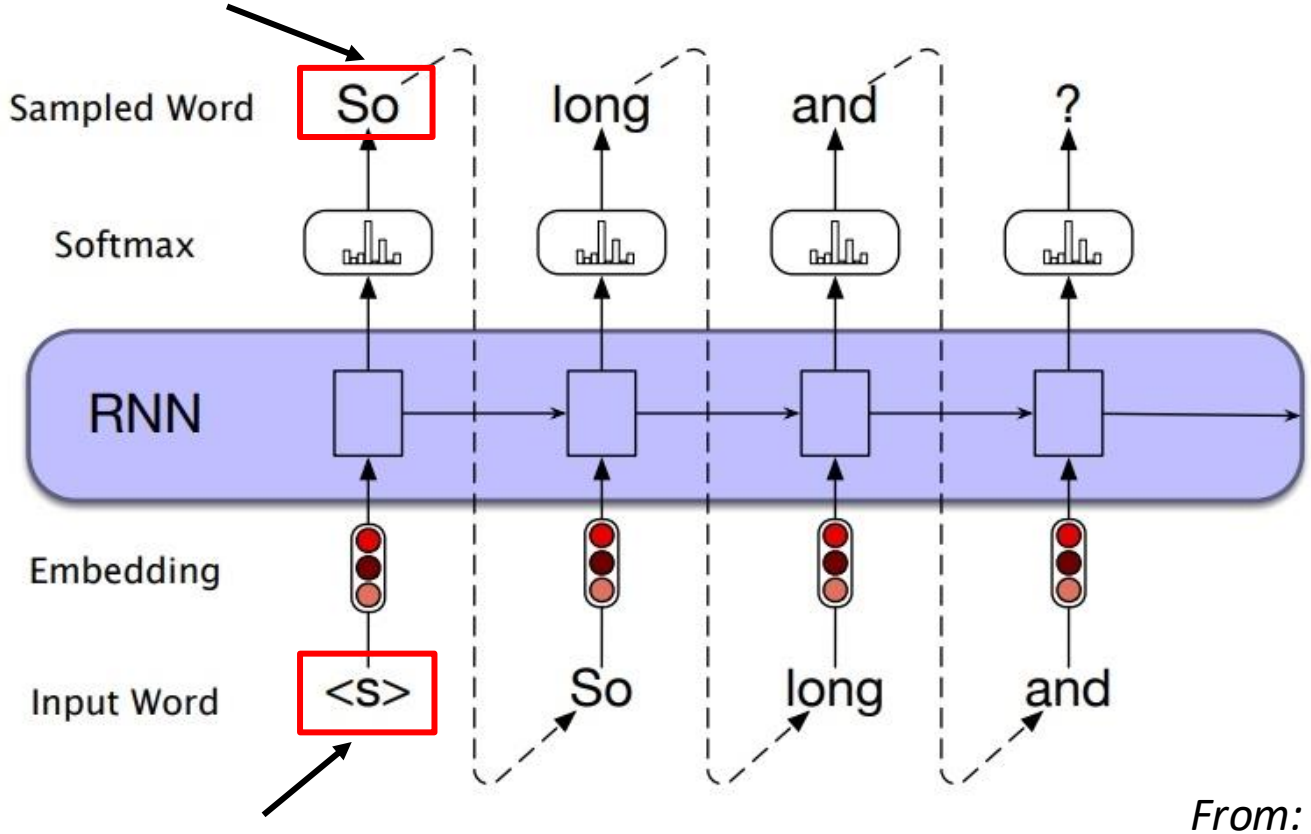
Each output of a sequential model depends on the hidden state variable provided by the previous step.

RNN for Sequence Generation

1. **Input Encoding:**
   - The input sequence is fed into the RNN one element at a time.
2. **Hidden State Update:**
   - The RNN updates its hidden state at each time step based on the current input element and the previous hidden state.
3. **Output Generation:**
   - Using the hidden states at each time step, the RNN generates the output sequence one element at a time.
4. **Iteration:**
   - Steps 2-3 are iterated until the desired length of the output sequence is reached or until a specific termination condition is met (e.g., generating an end-of-sequence token).

RNN for Sequence Generation

Predict the next token.

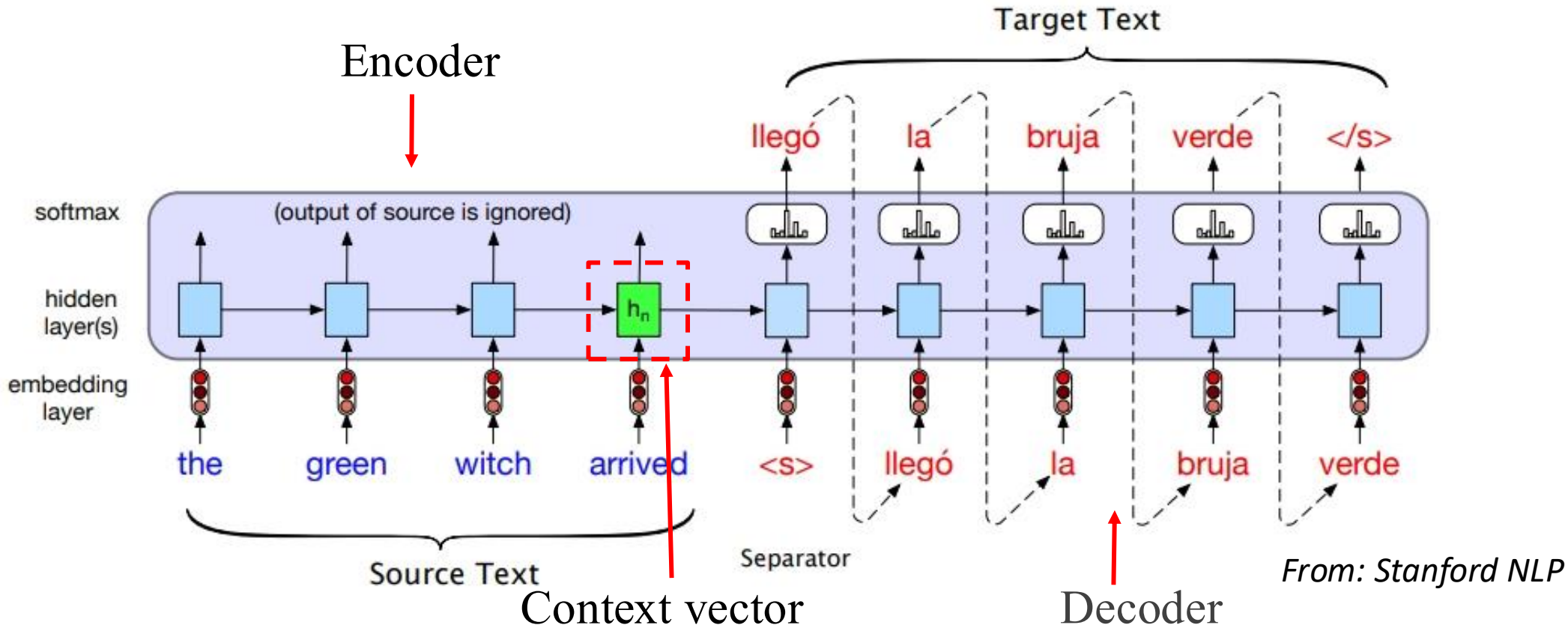


The sequence generation begin with a special token.

From: Stanford NLP

RNN for Machine Translation

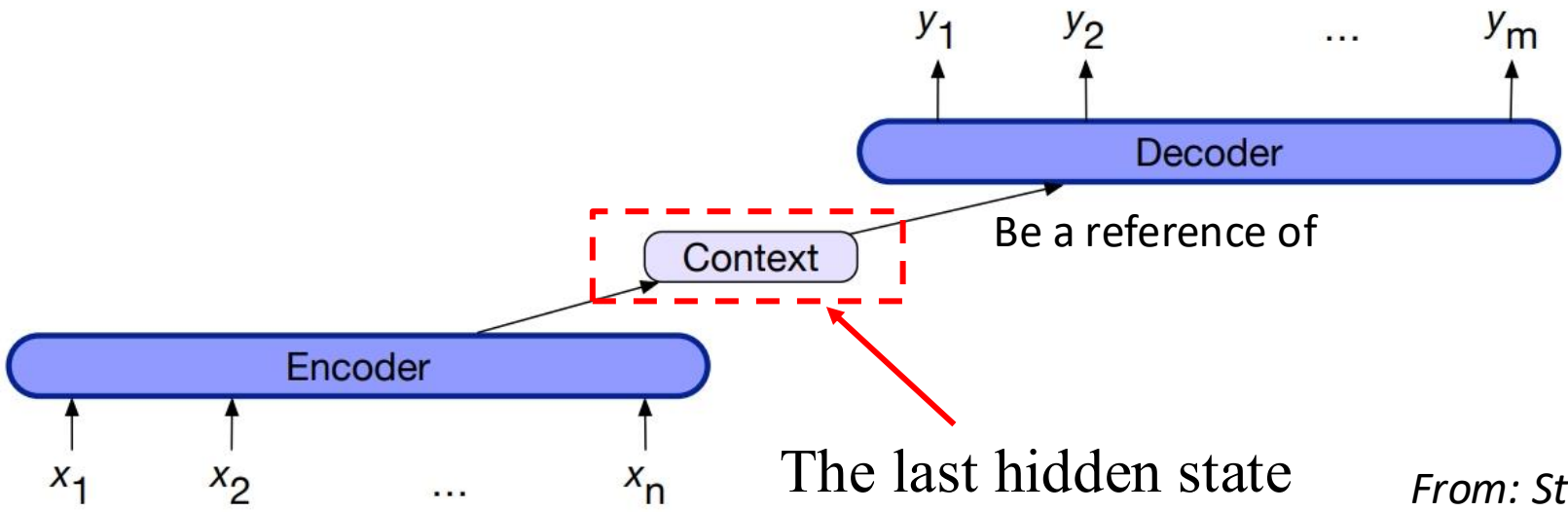
- Machine translation is a classic application of encoder-decoder architecture.



From: Stanford NLP

RNN for Machine Translation

- The context vector is the last hidden state of the encoder and it contains all the information of the input.



From: Stanford NLP

Encoder-decoder RNN

Encoder-decoder networks consist of three conceptual components:

- **An encoder** that accepts an input sequence, $x_{1:n}$, and generates a corresponding sequence of *contextualized representations*, $h_{1:n}$.
- **A context vector**, which is a function of $h_{1:n}$, and conveys the essence of the input to the decoder.
- **A decoder**, which accepts the context vector as input and generates an arbitrary length sequence of hidden states $h_{1:m}$, from which a corresponding sequence of output states $y_{1:m}$, can be obtained.

Gradient Vanishing/Exploding Problem of RNN

- During backpropagation, gradients are computed with respect to the hidden states through the loss function at each time step.

$$y_t = g(Vh_t)$$

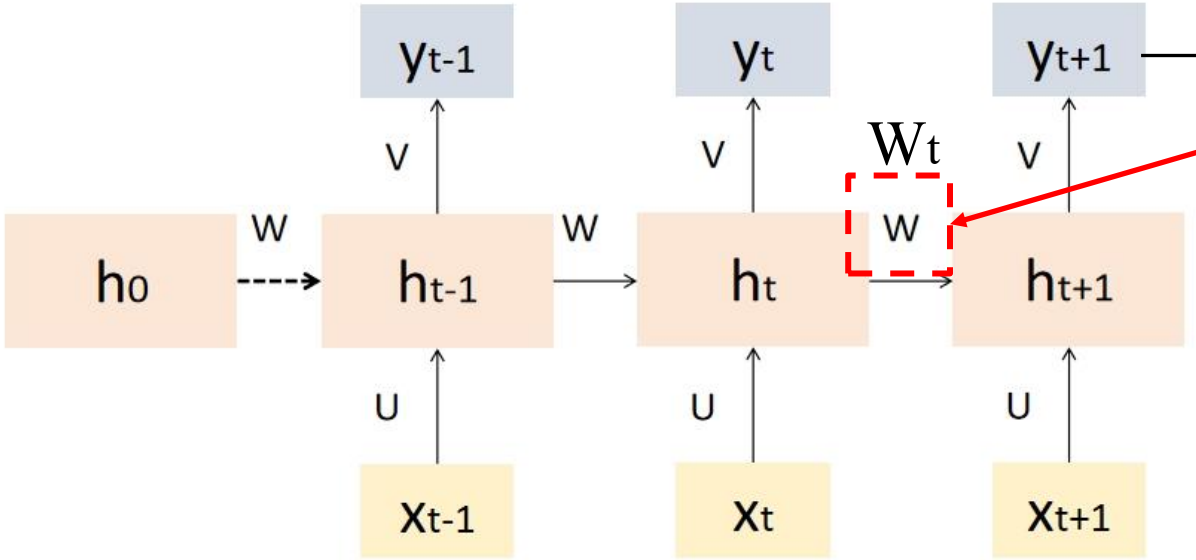
$$h_t = f(Ux_t + Wh_{t-1})$$

- Compute the gradient (use the parameter W as an example):

$$\frac{\partial L}{\partial W} = \sum_{t=1}^{T} \frac{\partial L}{\partial h_T} \cdot \frac{\partial h_T}{\partial h_{T-1}} \cdots \frac{\partial h_t}{\partial W}$$

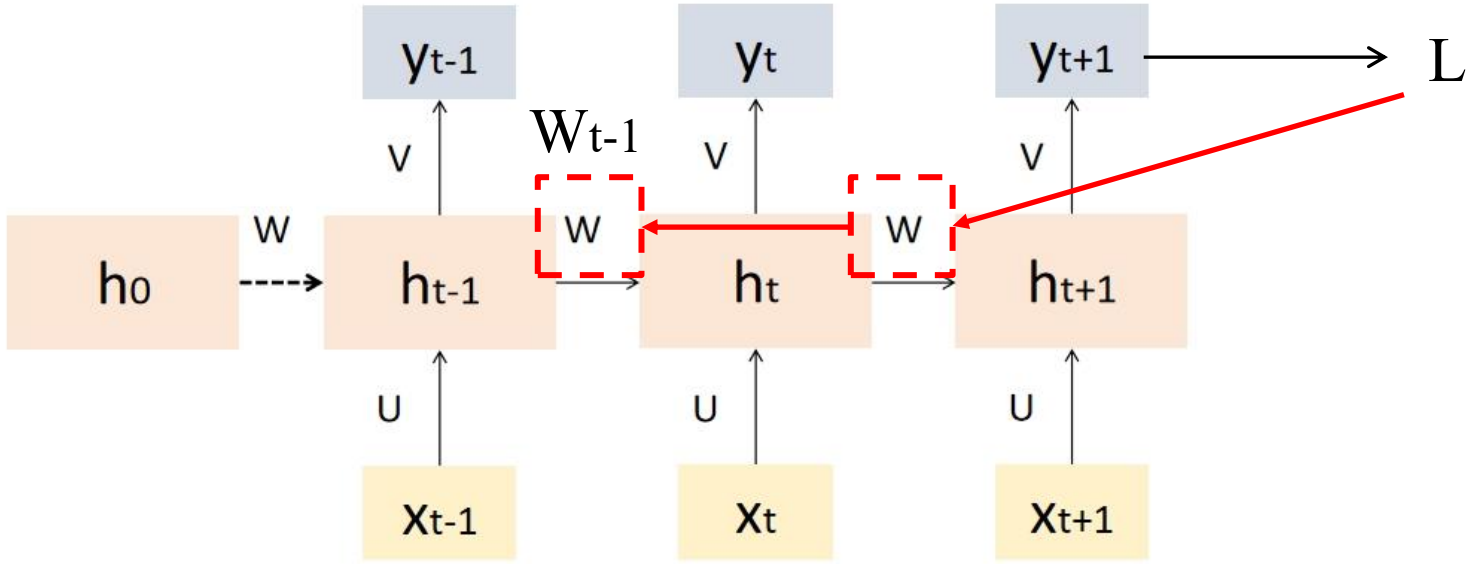
The further back in time an input is, the more factors it needs to be multiplied by.

Gradient Vanishing/Exploding Problem of RNN



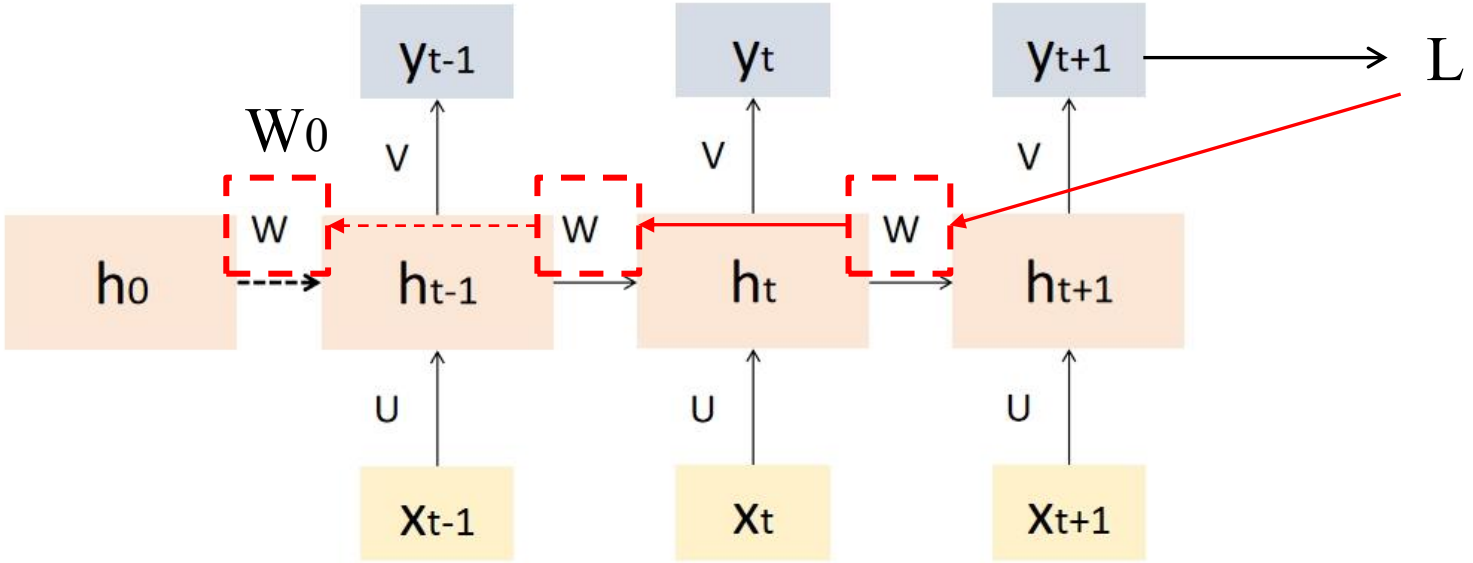
$$\nabla RNN(W_t) = \frac{\partial L}{\partial W_t}$$

Gradient Vanishing/Exploding Problem of RNN



$$\nabla RNN(W_{t-1}) = \frac{\partial L}{\partial W_{t-1}} = \frac{\partial L}{\partial W_t}\frac{\partial W_t}{\partial W_{t-1}}$$

Gradient Vanishing/Exploding Problem of RNN

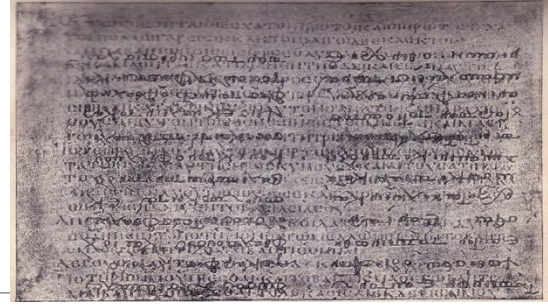


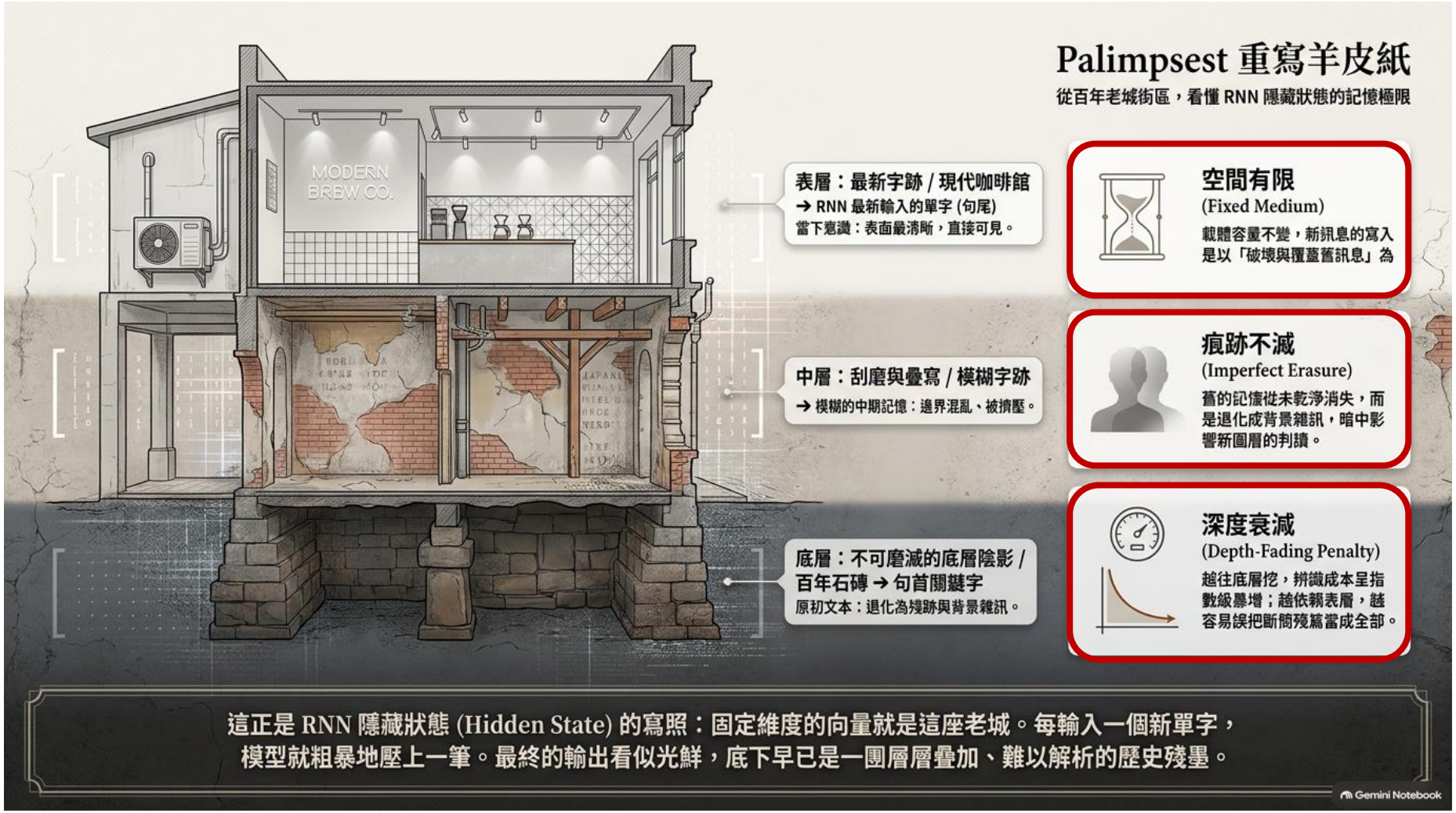
$$\nabla RNN(W_0) = \frac{\partial L}{\partial W_0} = \frac{\partial L}{\partial W_t}\frac{\partial W_t}{\partial W_{t-1}}\frac{\partial W_{t-1}}{\partial W_{t-2}}\cdots\frac{\partial W_1}{\partial W_0}$$

If these terms <1, the gradient decays exponentially (gradient vanishing)

If these terms >1, the gradient grows rapidly (gradient exploding)

Palimpsest (重寫羊皮紙)





Gradient Vanishing Problem of RNN

Gradient vanishing results in ：

- **Information Loss:**

  As the number of time steps increases, gradient vanishing prevents RNNs from retaining important information from earlier time steps.

- **Training Instability:**

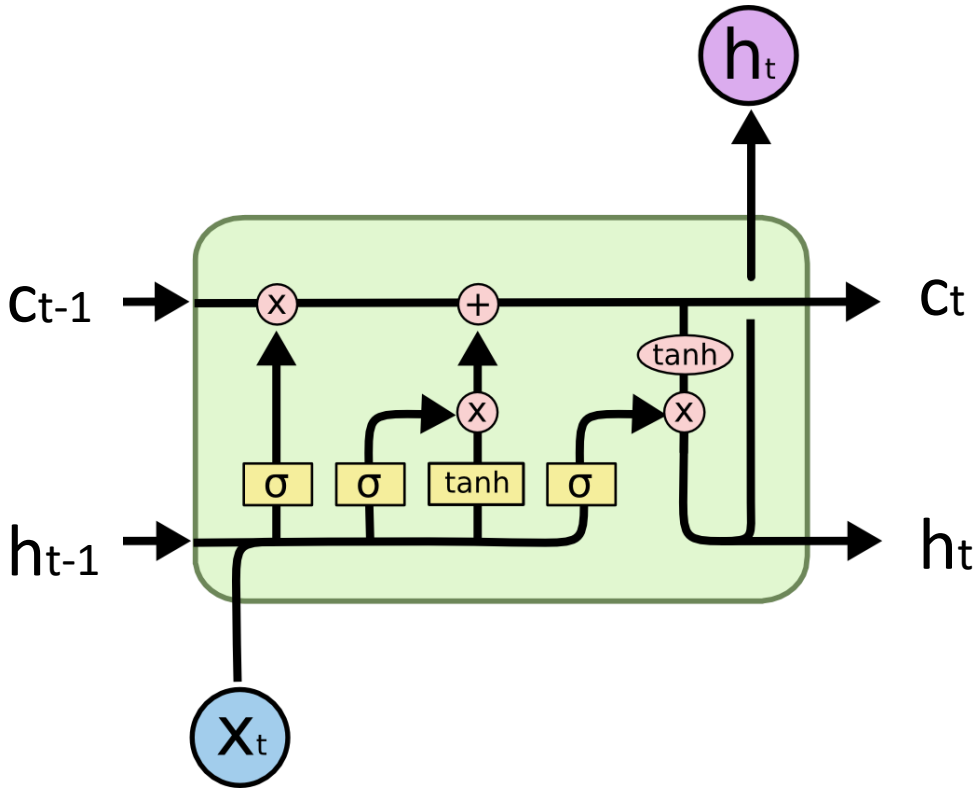
  With the disappearance of gradients in early time steps, the update of network parameters becomes slow or even stagnates.

- **Difficulty in Generating Long Sequences:**

  RNNs perform poorly in generating long sequences. Generated sequences may become incoherent or meaningless.

Long Short-Term Memory

Long Short-Term Memory (LSTM) is proposed to solve Gradient Vanishing problem of RNNs.



$$R_t = \sigma_r(W_r x_t + U_r h_{t-1} + b_r) \quad \text{Forget gate}$$

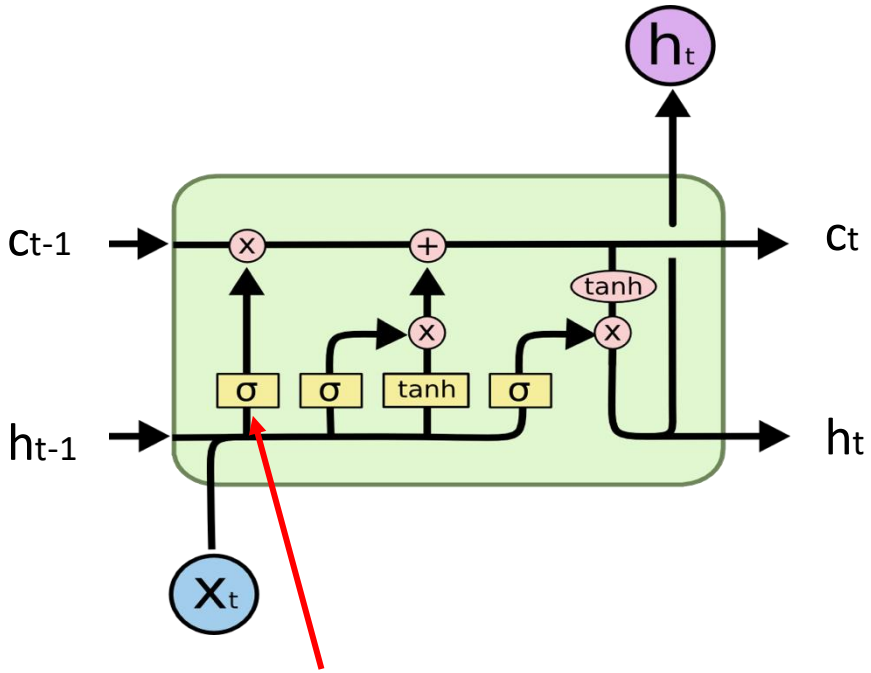
$$K_t = \sigma_r(W_k x_t + U_k h_{t-1} + b_k) \quad \text{Input gate}$$

$$V_t = \sigma_r(W_v x_t + U_v h_{t-1} + b_v) \quad \text{Output gate}$$

$$c_t = R_t \cdot c_{t-1} + K_t \cdot \sigma_c(W_c x_t + U_c h_{t-1} + b_c)$$

$$h_t = V_t \sigma_h(c_t)$$

Long Short-Term Memory

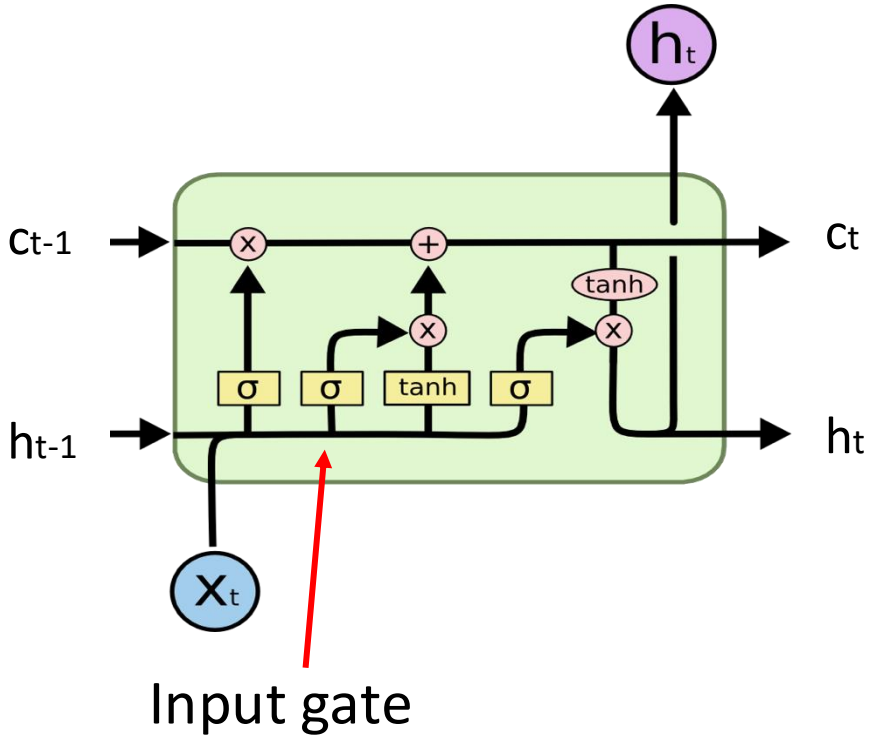


Forget gate

$$R_t = \sigma_r(W_r x_t + U_r h_{t-1} + b_r)$$

- **Forget Gate:** It uses a sigmoid function to determines which information from past memory should be forgotten or discarded.

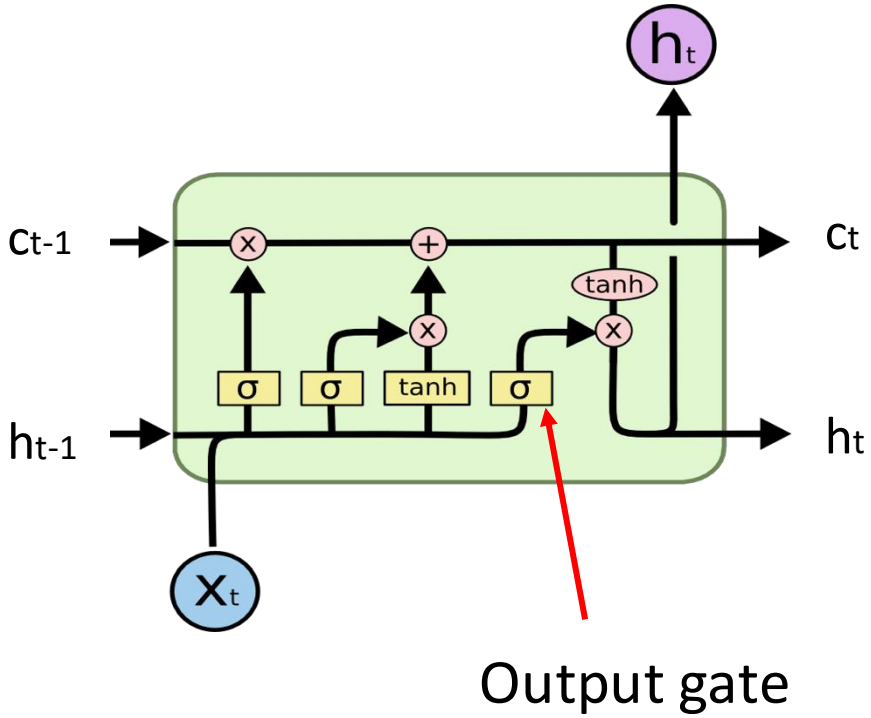
Long Short-Term Memory



$$K_t = \sigma_r(W_k x_t + U_k h_{t-1} + b_k)$$

- **Input Gate:** It uses a sigmoid function to determine which information should be added to the cell state.

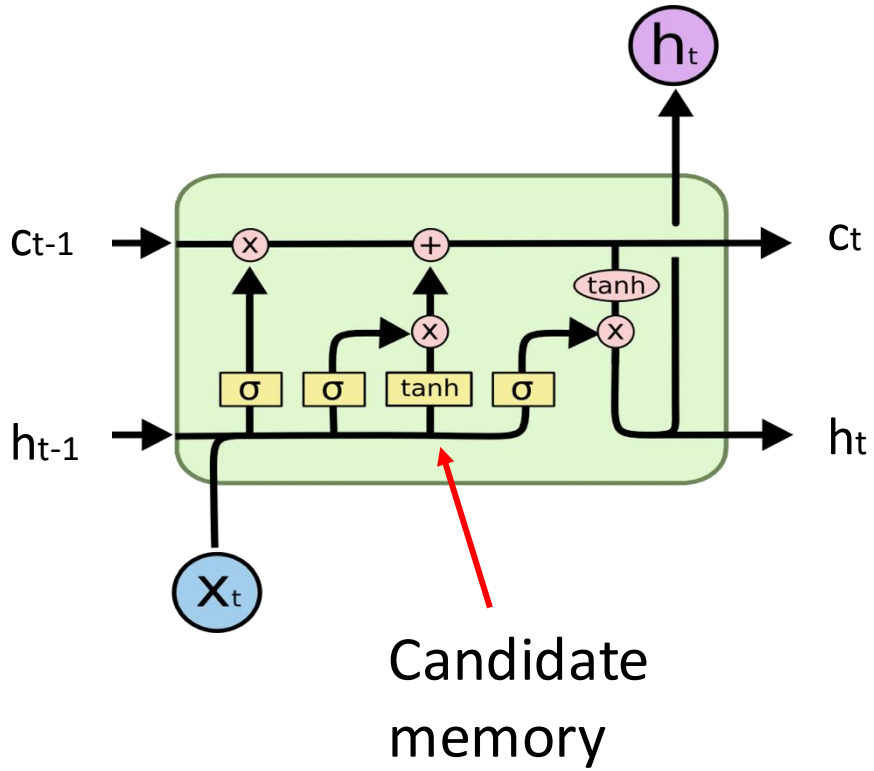
Long Short-Term Memory



$$V_t = \sigma_r(W_v x_t + U_v h_{t-1} + b_v)$$

- **Output Gate:** The output gate controls the flow of information from the cell state to the hidden state.

Long Short-Term Memory



$$c_t = R_t \cdot c_{t-1} + K_t \cdot \sigma_c(W_c x_t + U_c h_{t-1} + b_c)$$

$$h_t = V_t \sigma_h(c_t)$$

- **Candidate Memory:** The candidate memory represents potential new information that could be added to the cell state. It is generated by a tanh which is relatively closer to binary than sigmoid.

How does LSTM work?

Summary in plain words:
- **Forget gate** → drop "chasing" details.
- **Input gate** → add "climbing" action.
- **Candidate memory** → encode "climb + object."
- **Output gate** → release "tree" at the right time.

"*The cat chased the mouse, and then it climbed a tree*."



LSTM gate interpretations (language perspective)

Forget Gate

- When the model reaches "climbed a tree," it **reduces the importance of older context like "chased the mouse"** because the focus shifts to a new action.
- **Meaning**: drop past context that is less relevant for the current prediction.

Input Gate

- At "climbed," the model needs to **store this new action (climb) into the memory** since it defines what happens next.
- **Meaning**: decide which new information is worth adding to memory.

Candidate Memory

- Here, the candidate representation encodes the semantics of "climb + tree."
- **Meaning**: generate potential new content that could be added to the cell state.

Output Gate

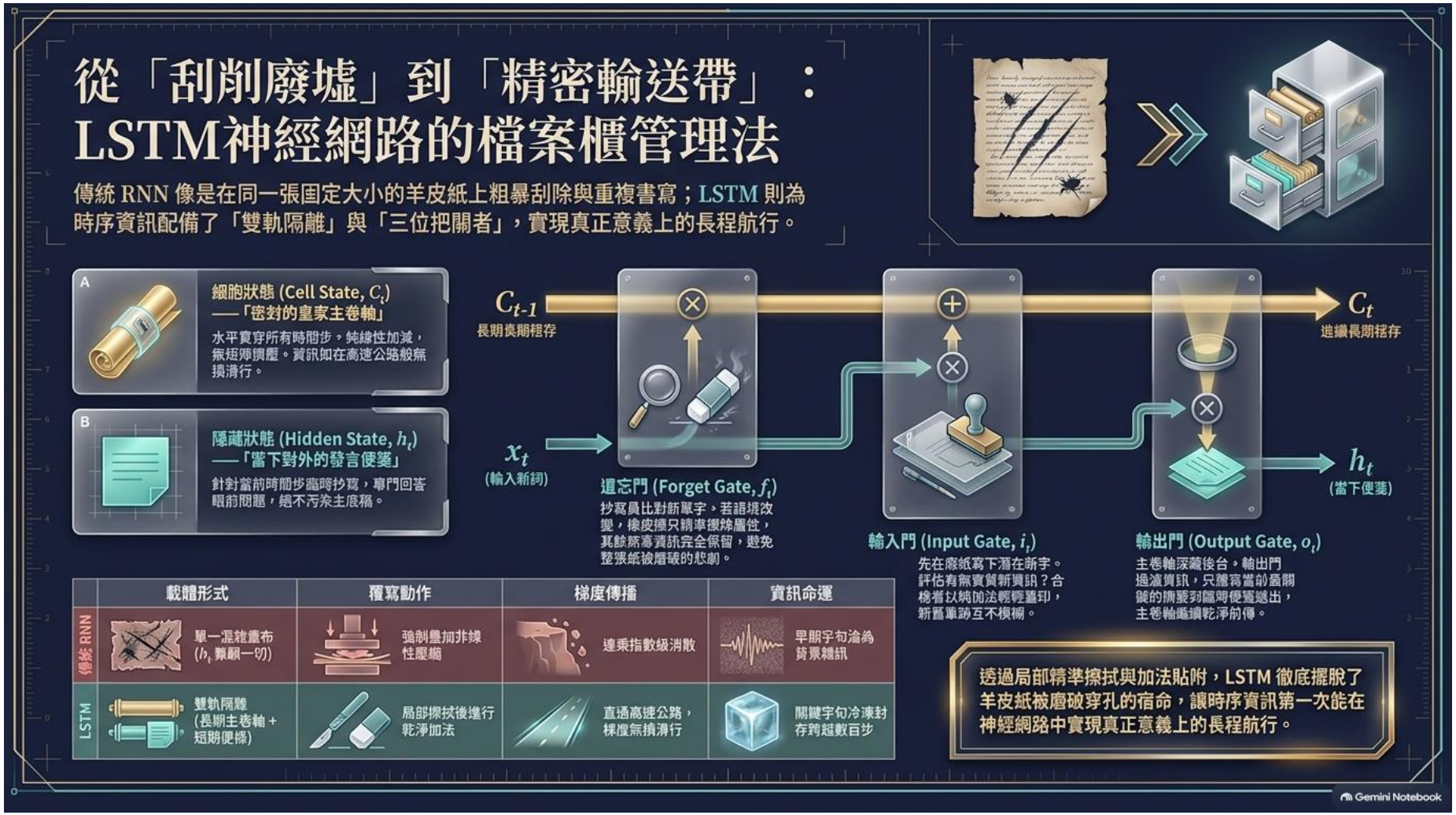
- When producing the word "tree," the model **selects from memory the relevant semantic content (the location being climbed)** to expose as output.
- **Meaning**: choose which part of the memory should influence the current word generation.

Long Short-Term Memory

LSTM can partially avoid the vanishing gradient problem due to：

- gating mechanisms & memory cells.
  - These components enable LSTM to **selectively retain or discard information** over time。
  - It maintains stable gradient flow during training and capture long-term dependencies more effectively compared to traditional RNNs.

Palimpsest (重寫羊皮紙) 2.0



Problems of time-series

Traditional sequential models, such as simple RNNs, suffer from the following issues:

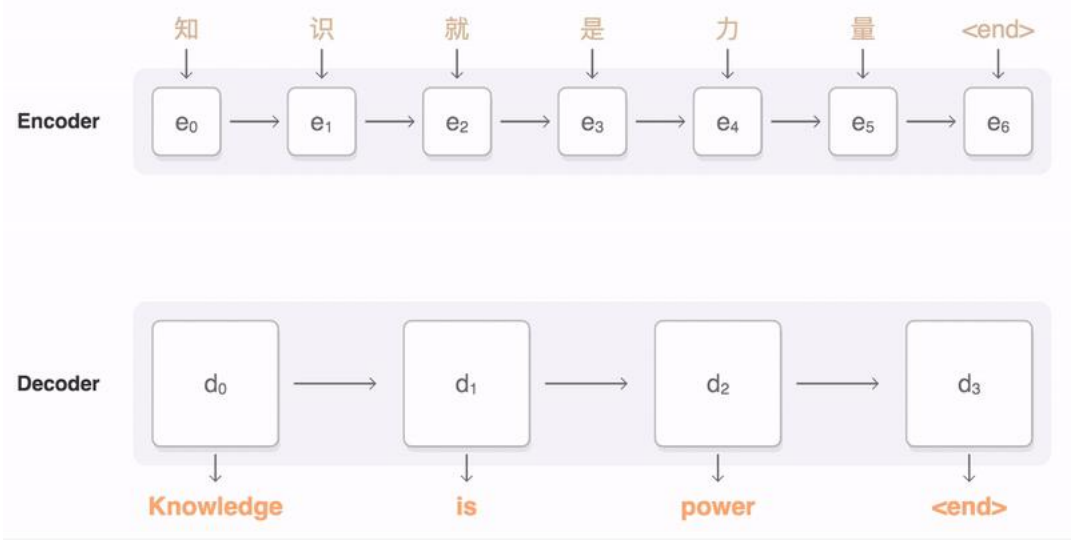
- **Vanishing/Exploding Gradients:** although LSTM alleviate this problem, it is still existed.
- **Difficulty in Parallelization:** Because sequential models rely on the previous time step's hidden state, they are challenging to parallelize effectively, limiting their training efficiency on large-scale datasets and the size-growth of language models.

Attention Mechanisms

The attention mechanism solve the gradient problems and provided a parallelizable solution of LMs.

Core Idea: To enable a model to **focus on the most relevant parts** of the input sequence when making predictions or generating outputs.

- **Attention**
  - Attention with RNNs
  - Attention without RNNs



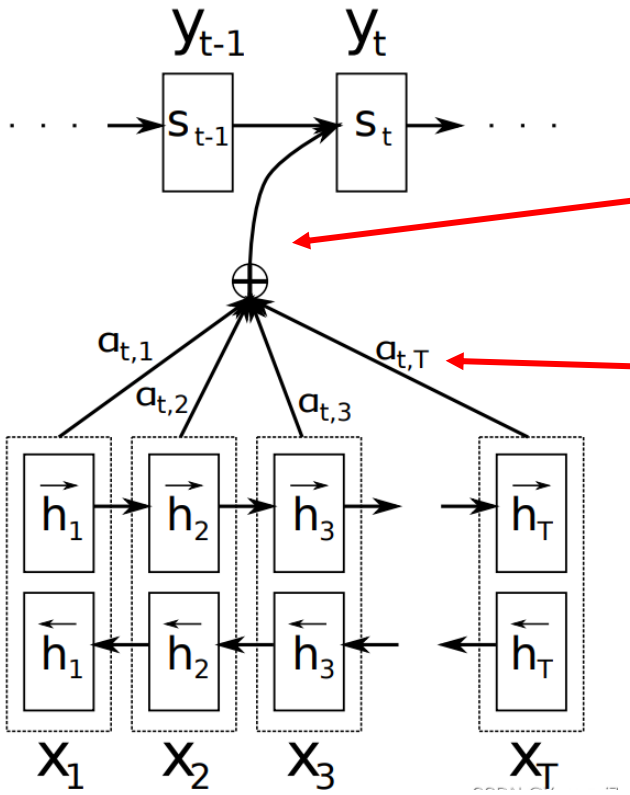
From Google Seq2seq

https://arxiv.org/abs/1409.0473, also in ICLR 2015

(additive attention)

Attention

At the beginning the attention is associated with RNNs.



From: Attention 2015

$$c_i = \sum_{j=1}^{T_x} \alpha_{ij} h_j$$ Weighted summarize

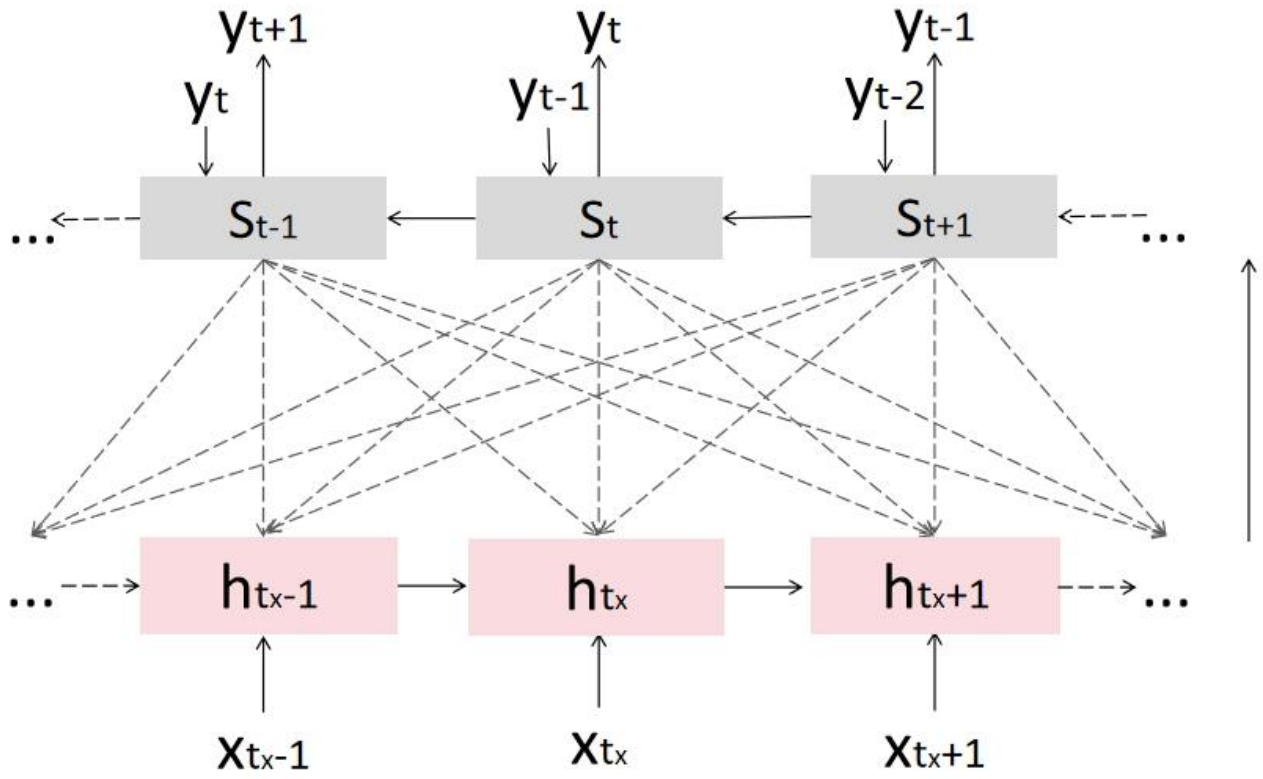
$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k=1}^{T_x} \exp(e_{ik})}$$

Where $e_{ij} = a(s_{i-1}, h_j)$

a is an alignment model which scores how well the inputs around position j and the output at position i match

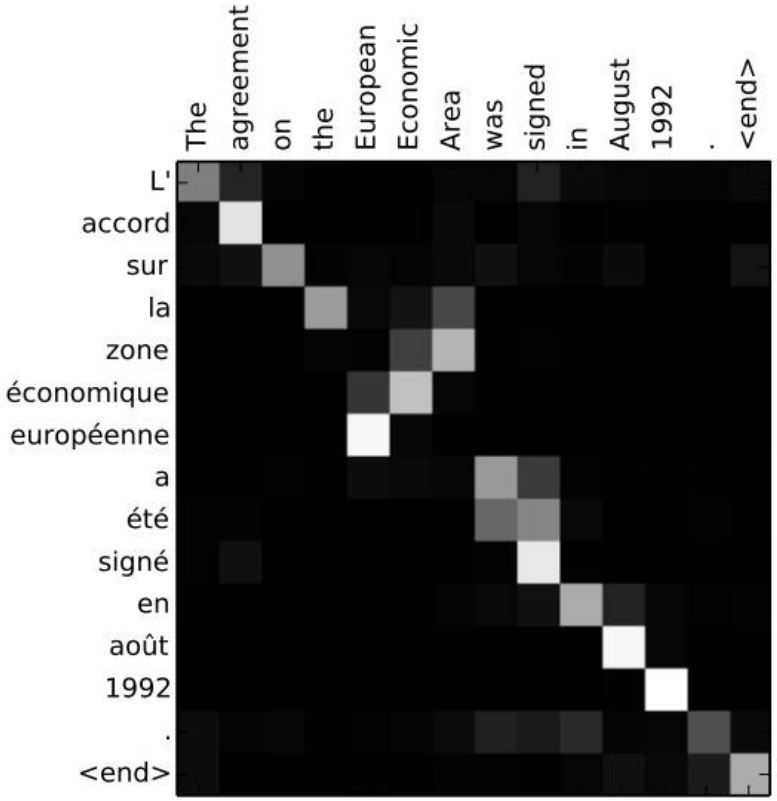
Attention

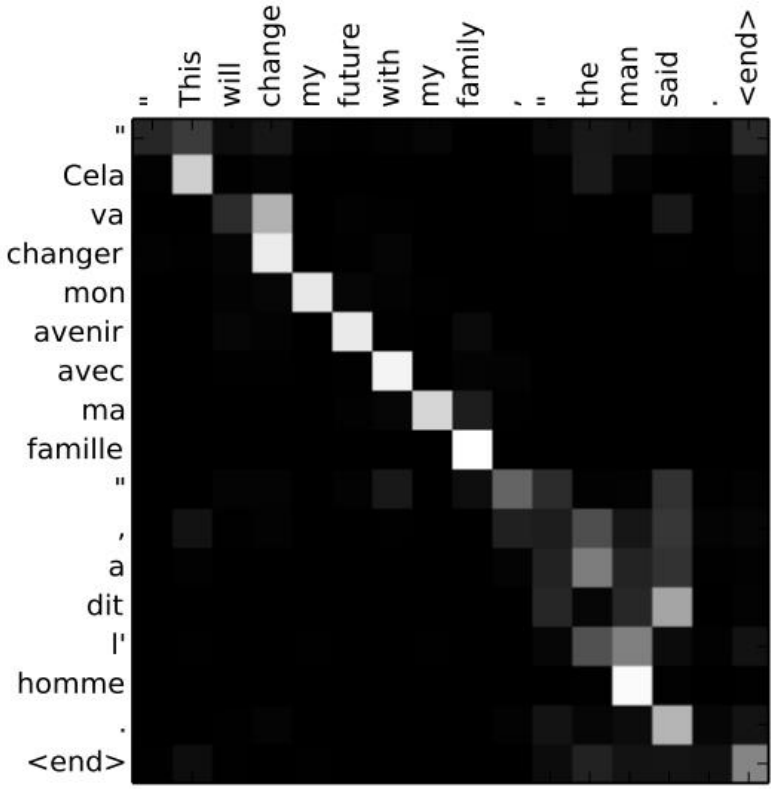
When computing the new hidden state s_t, attention mechanism compute attention scores with all the input tokens (from a bidirectional RNN).



Attention

An example of machine translation.





From: Attention 2015

Attention without RNNs

It has been proposed to eliminate the RNN component due to:

- The primary function of the RNN is to extract and process sequential features. However, this functionality can be achieved using simpler methods, leading to **reduced computational complexity**.
- RNNs are **not parallelizable**, meaning they cannot efficiently process multiple input sequences simultaneously, which limits their scalability and efficiency in large-scale applications.

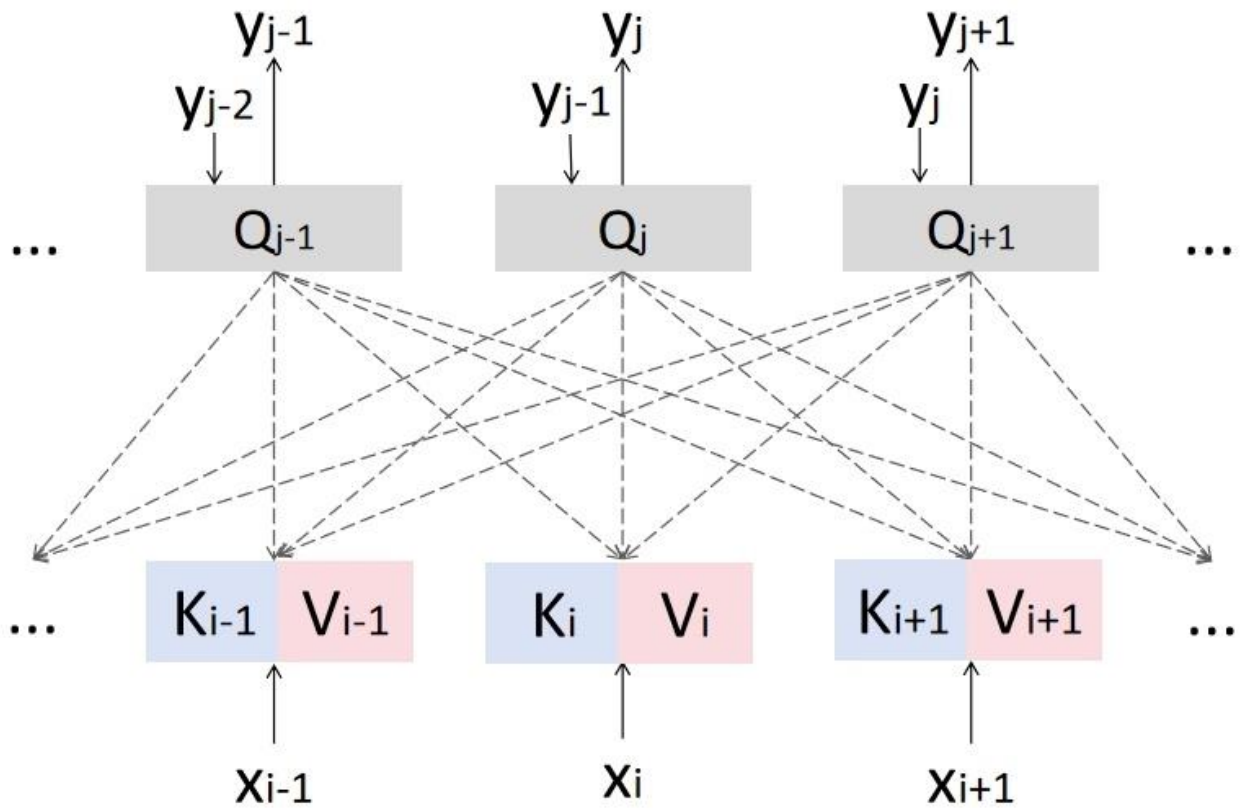
Attention without RNNs

$x, y \in \mathbb{R}^{d_{model} \times N}$

$W_Q, W_K \in \mathbb{R}^{d_k \times d_{model}}$

$W_V \in \mathbb{R}^{d_v \times d_{model}}$

N *is text length,* d_{model} *is the size of embedding*



Learnable weight matrix

$$Q = W_Q \cdot y$$
$$K = W_K \cdot x$$
$$V = W_V \cdot x$$

output

input

$Q^T K \in \mathbb{R}^{N \times N}$

Attention score:

$$\alpha_i = Softmax(\frac{Q^\top K}{\sqrt{d}})$$

$$y_{output} = \sum_{i=1}^{N} \alpha_i v_i$$

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

eq in the original paper

Palimpsest (重寫羊皮紙) 3.0

徹底廢黜時間輸送帶：Transformer 的空間革命

從「排隊傳抄」到「全景攤平」，語言模型如何透過 QKV 機制將時間序列轉化為幾何空間。

Architectural Evolution Matrix

1 初代 RNN (單張羊皮紙)



- 機制：X_t 暴力塗改覆寫。
- 痛點：舊字成雜訊，導致記憶模糊崩潰。

2 二代 LSTM (密封長卷軸)



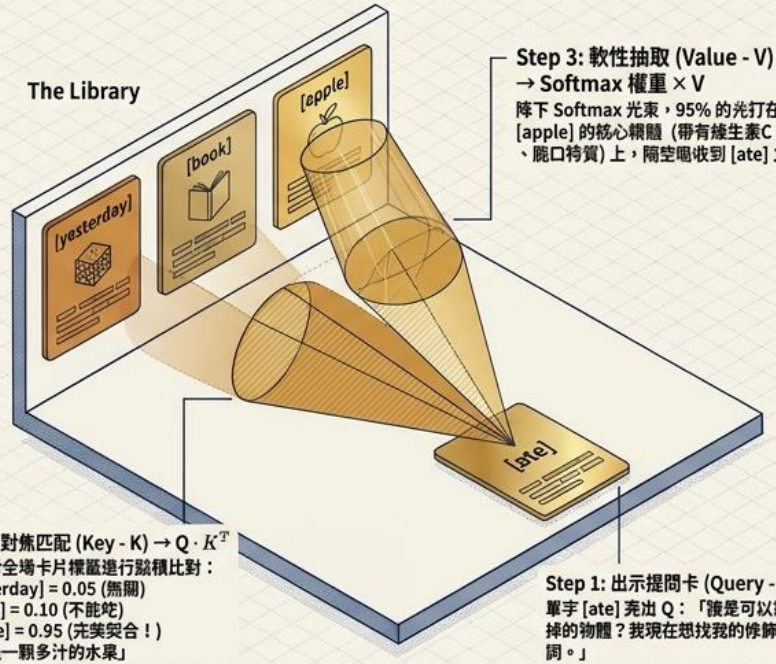
- 機制：主卷軸 C_t 透過過濾閘門進行局部加法。
- 痛點：序列鎖鏈。第 100 頁必須滑過 99 個抄寫員無法平行運算（後面必須等前面）。

3 現代 Transformer (卡片圖書館)



- 機制：全域攤平，每字化為獨立圖書卡片。
- 突破：距離歸零，任意跨步瞬間加權抓取。

The Core Engine (QKV 三聯檢索卡)



Step 3: 軟性抽取 (Value - V) → Softmax 權重 × V

降下 Softmax 光束，95% 的光打在 [apple] 的核心輻射（帶有維生素C、甜味、脆口特質）上，隔空吸收到 [ate] 之中。

Step 2: 對焦匹配 (Key - K) → $Q \cdot K^T$

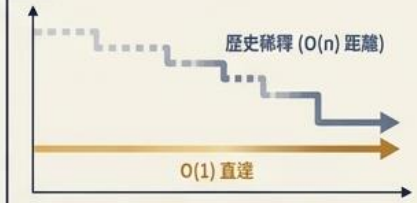
瞬間掃射全場卡片標籤進行點積比對：
[yesterday] = 0.05 (無關)
[book] = 0.10 (不能吃)
[apple] = 0.95 (完美契合！)
「我是一顆多汁的水果」

Step 1: 出示提問卡 (Query - Q)

單字 [ate] 亮出 Q：「誰是可以被我吃掉的物體？我現在想找我的修飾詞成受詞。」

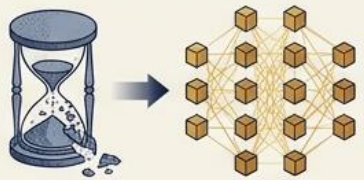
The Two Century Problems Solved

3A 告別歷史稀釋 (O(1) 恆定記憶)



不管兩張卡片隔了多遠，提問 Q 與標籤 K 皆是一步直達 (O(1) 距離)。第 1 個字與第 1,000 個字對話的清晰度，跟相鄰兩字完全一模一樣。

3B 撕掉「時間束縛」(平行化運算)



$$\text{Attention}(Q,K,V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

整本 5,000 字的書無需排隊，GPU 在同一個時鐘週期內，透過一次矩陣乘法，瞬間完成全場所卡片兩兩之間的比對。

BOTTOM FOOTER MODULE

哲學躍遷： 處理時間序列，不一定需要「時間的順序」。把時間攤平成幾何空間，賦予每個單字直接去圖書館貨架檢索他人的權力，語言模型才真正走出了記憶衰褪的迷宮。

Gemini Notebook

Why $\sqrt{d}$

The model's parameters should undergo normalization, ensuring their average is 0 and variance is 1.

Assume that q_i and k_i are random variable with average 0 and variance 1:

$$
\begin{aligned}
E(q_i k_i) &= E(q_i)E(k_i) = 0 \\
Var(q_i k_i) &= E(q_i^2 k_i^2) - E(q_i k_i)^2 \\
&= E(q_i^2 - 0^2)E(k_i^2 - 0^2) \\
&= E(q_i^2 - E(q_i)^2)E(k_i^2 - E(k_i)^2) \\
&= Var(q_i)Var(k_i) = 1
\end{aligned}
$$

Why $\sqrt{d}$

However, after we multiply Q and K, the variance becomes d:

$$Var(Q^\top K) = Var(\sum_{i=0}^{d} q_i k_i) = d \cdot 1 = d$$

So it should be divided by $\sqrt{d}$:

$$Var(\frac{Q^\top K}{\sqrt{d}}) = \frac{d}{(\sqrt{d})^2} = 1$$

Summary

Sequential Models:

RNN: Recurrent Neural Network, the fundamental NN in NLP tasks.

LSTM: A modification of RNNs designed to alleviate gradient issues.

Attentions:

Attention with RNN: Avoid gradient vanishing in RNNs.

Attention without RNN: Parallelizable and simpler.

Reference

Stanford NLP:

- https://web.stanford.edu/~jurafsky/slp3/

Attention 2015:

- Bahdanau, D., Cho, K., & Bengio, Y. (2014). Neural machine translation by jointly learning to align and translate. arXiv preprint arXiv:1409.0473.